

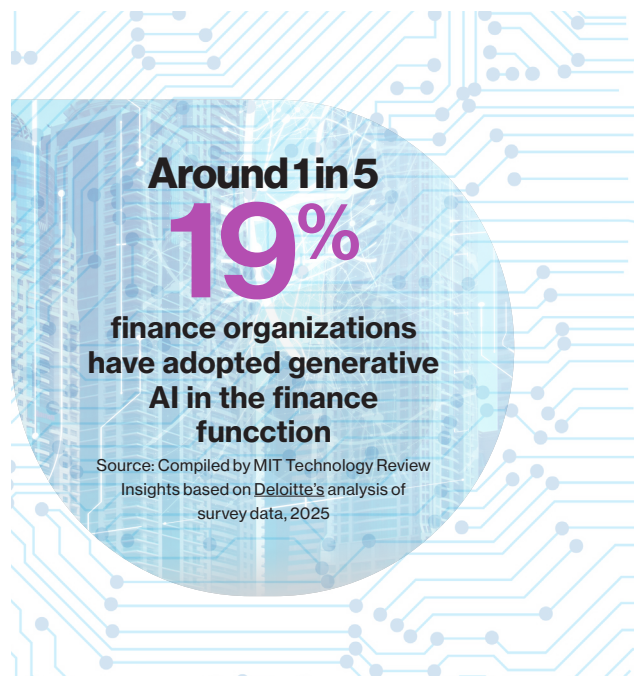
CFOs are experimenting with AI use cases to free up capacity for business-critical work.

Partnering with generative AI in the finance function



Generative AI has the potential to transform the finance function. By taking on some of the more mundane tasks that can occupy a lot of time, generative AI tools can help free up capacity for more high-value strategic work. For chief financial officers, this could mean spending more time and energy on proactively advising the business on financial strategy as organizations around the world continue to weather ongoing geopolitical and financial uncertainty.

CFOs can use large language models (LLMs) and generative AI tools to support everyday tasks like generating quarterly reports, communicating with investors, and formulating strategic summaries, says



Key takeaways

- 1 **Generative AI can support financial professionals with rote or repetitive tasks, freeing their time for more proactive or strategic work for the business.**
- 2 **Although the technology hasn't reached maturity and cybersecurity and hallucination challenges persist, most finance organizations are increasing their investment in generative AI, attracted by its potential for increased productivity and process optimization.**
- 3 **The future of generative AI for CFOs could lie in engineering "capability automation," where tasks or processes are combined across multiple steps.**

Andrew W. Lo, Charles E. and Susan T. Harris professor and director of the Laboratory for Financial Engineering at the MIT Sloan School of Management. "LLMs can't replace the CFO by any means, but they can take a lot of the drudgery out of the role by providing first drafts of documents that summarize key issues and outline strategic priorities."

Generative AI is also showing promise in functions like treasury, with use cases including cash, revenue, and liquidity forecasting and management, as well as automating contracts and investment analysis. However, challenges still remain for generative AI to contribute to forecasting due to the mathematical limitations of LLMs. Regardless, [Deloitte's analysis](#) of its [2024 State of Generative AI in the Enterprise survey](#) found that one-fifth (19%) of finance organizations have already adopted generative AI in the finance function.

Despite return on generative AI investments in finance functions being 8 points below expectations so far for surveyed organizations (see Figure 1), some finance departments appear to be moving ahead with investments. Deloitte's fourth-quarter [2024 North American CFO](#)

Signals survey found that 46% of CFOs who responded expect deployment or spend on generative AI in finance to increase in the next 12 months (see Figure 2). Respondents cite the technology's potential to help control costs through self-service and automation and free up workers for higher-level, higher-productivity tasks as some of the top benefits of the technology.

“Companies have used AI on the customer-facing side of the house for a long time, but in finance, employees are still creating documents and presentations and emailing them around,” says Robyn Peters, principal in finance transformation at Deloitte Consulting LLP. “Largely, the human-centric experience that customers expect from brands in retail, transportation, and hospitality haven’t been pulled through to the finance organization. And there’s no reason we cannot do that – and, in fact, AI makes it a lot easier to do.”

If CFOs think they can just sit by for the next five years and watch how AI evolves, they may lose out to more nimble competitors that are actively experimenting in the space. Future finance professionals are growing up using generative AI tools too. CFOs should consider reimagining what it looks like to be a successful finance professional, in collaboration with AI.

The ROI of AI in finance

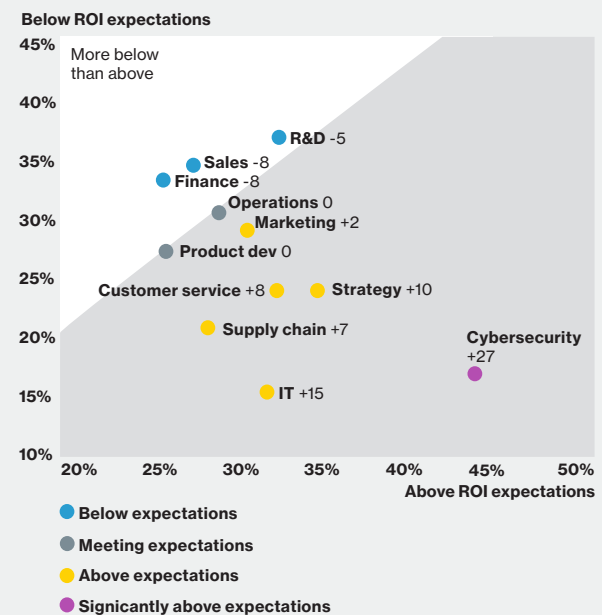
As a business grows in revenue, CFOs typically want to avoid growing functions like HR, finance, and marketing at the same pace. There’s a huge opportunity with AI to “flatten the cost curve” and do more with less, says Peters. “Finance organizations can have high headcounts and can be somewhat inefficient,” agrees James Glover, a principal in finance transformation at Deloitte Consulting LLP. “Back-office functions are always under pressure from a cost and efficiency perspective. Companies ask if they can do the same amount of work with fewer people or if they can do more work with the same amount of people?”

That means making the finance function more productive. Finance professionals often work with a host of different systems for activities including accounting, forecasting, and customer relationship management (CRM). Executives can use agentic AI to pull data from many sources and systems into a single source to help them make decisions more quickly, organize complex workflows, and even coordinate tasks with other agents.

“AI is the new UI,” says Peters. “Professionals don’t have to log into seven different systems to do their job. They can have a ‘single pane of glass’ – one experience layer – that is enabled through natural language and gives them information from all these underlying systems.”

Some finance organizations are already moving ahead, building innovations on the customer-facing side in areas like dynamically adjusted and hyper-personalized portfolio management in financial services. “We are still in the early- to mid-stages of generative AI maturity, with internal processes like financial reporting and financial planning and analysis (FP&A) further along in adoption compared to mission-critical, externally regulated areas such as portfolio management and investment advisory,” says the head of finance at a large international investment firm. “While we’ve made progress in inward-looking functions, the sandbox environment and manual reviews remain essential in regulated services due to compliance requirements.”

Figure 1: Generative AI is not meeting ROI expectations of those surveyed in the finance function



The numbers with each function indicate the difference between the % above expectation and the % below expectations.

Source: Compiled by MIT Technology Review Insights, based on data from Deloitte, 2025

Overcoming the challenges of AI adoption in finance

Top cybersecurity risks for executives include malware, ransomware, phishing, and data loss threats, according to the [fourth edition of Deloitte's Global Future of Cyber Survey](#). Generative AI can increase the effectiveness and sophistication of these kinds of threats, including enhancing deepfakes for financial fraud, for example. Deloitte's Center for Financial Services predicts that generative AI could enable [fraud losses](#) to reach \$40 billion in the US by 2027, up from \$12.3 billion in 2023, growing at a compound annual growth rate of 32%. In addition, many executives reported they are concerned about the propensity of AI models for [inaccuracy and bias](#) and their tendency to lack transparency and explainability about how they work. Without software add-ons that focus on numerical content, many LLMs also show an alarming ability to [hallucinate with math](#), for example.

Regulatory compliance and data security requirements demand heightened efforts to mitigate these risks through human oversight. According to [Deloitte's 2024 State of Generative AI in the Enterprise quarter four report](#), compliance with regulation is the top challenge for companies to develop and deploy generative AI. Some 38% of respondents cited this as a challenge in the fourth quarter of 2024, up from 28% in the first. Difficulty managing risks was the second biggest concern for respondents by the end of 2024.

"AI as a field is growing exponentially fast right now," says Lo. "We have to pick up our pace in thinking about how to address some of these concerns, because AI has outpaced the regulators. All stakeholders should come together and figure out a responsible approach to putting up guardrails so that we're protected while encouraging innovation at the same time."

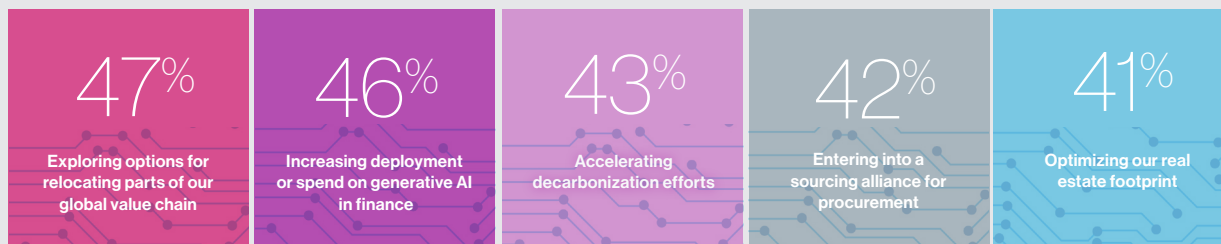
"Being a co-pilot to a CEO or business unit leader is what finance aspires to be. If AI can help take away the production effort, you can spend more time being strategic."

James Glover, Principal in Finance Transformation, Deloitte Consulting LLP

A big unknown for finance leaders, says Lo, is whether generative AI tools are able to satisfy fiduciary duty. "Can an AI system put its own interests above ours, or our interests above its own?" he says. "I'm not even sure AI knows what its own interests are, because that's something I don't think has ever been programmed into any kind of generative AI tool."

In the context of financial advice, one solution Lo and his colleagues are exploring involves [training AI systems to develop a greater awareness](#) of their duty to act in the best financial interests of humans. To properly train an LLM, they are using a large database of all securities violations tracked by the US Securities and Exchange Commission (SEC) and all the enforcement actions it has taken against actors who have violated their fiduciary duty. The thinking is that humans develop awareness of right and wrong through years of experience, even when they aren't exactly aware of what they know. So why can't machines do the same?

Figure 2: Nearly half of CFOs surveyed expect deployment or spend on generative AI to increase in the finance function in the next year



Source: Compiled by MIT Technology Review Insights, based on data from [Deloitte](#), 2025

After several years of experimentation, it appears many concerns are starting to be addressed through more accurate, secure, and responsible models and architectures, as well as turnkey solutions and the built-in features of existing finance platforms from major vendors. Model improvements are also allowing for greater explainability about exactly which data sources feed into a forecast.

“Overall familiarity is getting stronger among CFOs and it’s alleviating some of the previous barriers that we saw when everything felt new and somewhat scary,” says Peters. “After all, no CFO or chief accounting officer wants to have a material weakness from their auditor because they use generative AI to create some sort of financial statement.”

Effective AI adoption hinges on giving employees an “80% quality” start on the work that they do manually every day, says Peters. Consider the budget creation cycle. Most companies go through a budgeting process and then forecasts against what they see trending throughout the year. But that budgeting process is sometimes subject to bias, because as humans we sometimes prefer to under-promise and over-deliver, Peters says.

“AI as a field is growing exponentially fast right now and has outpaced the regulators. All stakeholders should come together and figure out a responsible approach to putting up guardrails so that we’re protected while encouraging innovation at the same time.”

Andrew W. Lo, Charles E. and Susan T. Harris Professor and Director of the Laboratory for Financial Engineering, MIT Sloan School of Management

Three use-case patterns in finance

James Glover, a principal at Deloitte Consulting LLP, identifies three ways of using AI that appear to be demonstrating the most buy-in among CFOs for the finance function:

Search, summarize, and extract

AI systems pull together a mountain of unstructured data, such as documents and calls related to investor relations. Employees can use an enterprise generative AI tool to analyze earnings transcripts, analyst reports, contracts, or accounting and tax policies. They can collect all the documents and files in a database, and employees can query the data with questions and create custom analyses without the time-consuming step of reading and analyzing all that material. Challenges to address include protecting sensitive proprietary data.

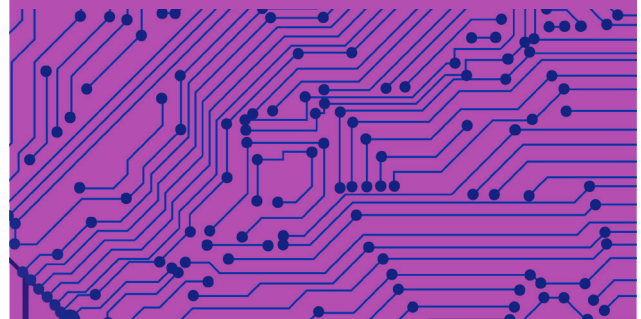
Talk to the data

Often finance organizations use an LLM-based AI chatbot to ask questions of structured data, such as, “What were the sales in my business unit for February 2024 compared with February 2023?” A global technology services firm removed the friction employees experience from searching multiple systems and reports when it deployed a similar solution. Challenges include gaining trust in the accuracy of the answers, especially those involving calculations and math.

Uncover data issues and make predictions

Traditional machine learning models have been around long before generative AI turbocharged prediction and forecasting. Some companies use AI to find anomalies in data and predict how to cut costs or fix errors. Another approach involves forecasting future demand, revenues, sales, and other sources of value. Challenges include explainability and transparency of reasoning and data sources.

The next phase could involve combining these use-case patterns with agentic AI to address capabilities across processes, functions, and workflows.



“While it is true that there are hurdles around things like data security and hallucinations, we shouldn’t forget that some of the hurdles already exist in how we do work right now.”

Robyn Peters, Principal in Finance Transformation, Deloitte Consulting LLP

When companies introduce machine learning and generative AI to these business processes, the degree of accuracy can actually rise, just by removing the human biases that exist in some finance processes today. “While it is true that there are hurdles around things like data security and hallucinations, we shouldn’t forget that some of the hurdles already exist in how we do work right now,” says Peters.

Reimagining the future of finance

The future of generative AI for CFOs could lie in engineering “capability automation,” or a combination of tasks and processes throughout multistep processes, functionalities, or workflows. “Now with agentic AI, companies can use multiple AI solutions to build task-based agents, process-based agents, or ultimately capability-based agents, depending on how you scale it,” says Glover. Humans will play important roles “in the

loop,” refining the way AI learns and gets better over time. Humans will also review proposed actions to make the final decisions.

Ultimately, AI has the potential to not only create and manage information, but to build capacity. “Finance has always had a mantra that it’s got to be a steward of the assets of the business,” says Glover. “But we also have to be a partner to the business and help the business plan, help the business pivot, help the business make good financially disciplined decisions.”

For many, the goal is to steer an organization’s capacity toward collaboration with leaders to effectively help them run the business. “Being a co-pilot to a CEO or business unit leader, that’s what finance aspires to be,” says Glover. “And if AI could help take away the production effort, time could be spent being more strategic.”



“Partnering with generative AI in the finance function” is an executive briefing paper by MIT Technology Review Insights. Virginia Wilson was the editor of this report, and Nicola Crepaldi was the publisher. MIT Technology Review Insights has independently collected and reported on all findings contained in this paper.

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